

Robotics & AI Club · Introduction

Welcome to Robotics!

What robots are, how they work, and your first build

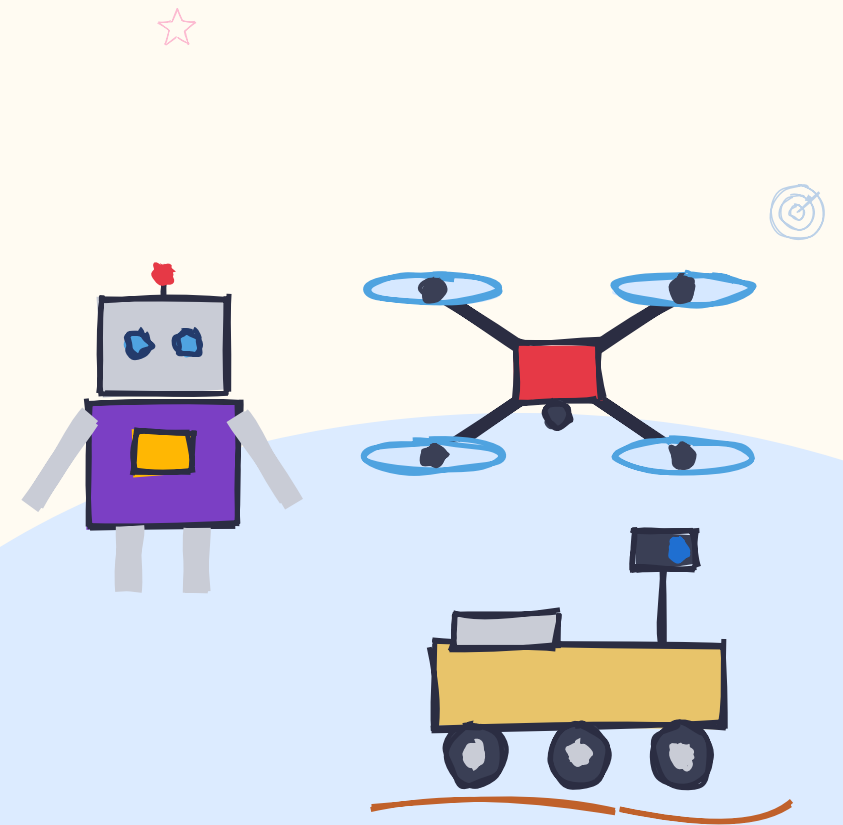
Sense, think, act

Parts of a robot

Arduino + ultrasonic sensor

Did you know?

There are more than 4 million robots working in factories around the world today.



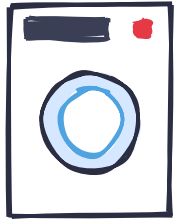


Quick quiz

Is it a robot? Thumbs up or down!

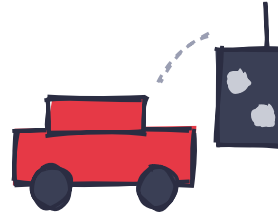


1



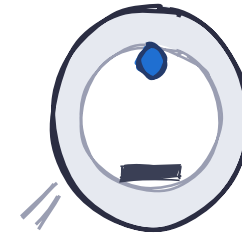
Washing machine

2



Remote-control car

3



Robot vacuum

4



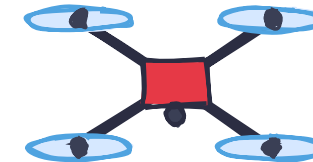
Calculator

5



Self-driving car

6



Delivery drone

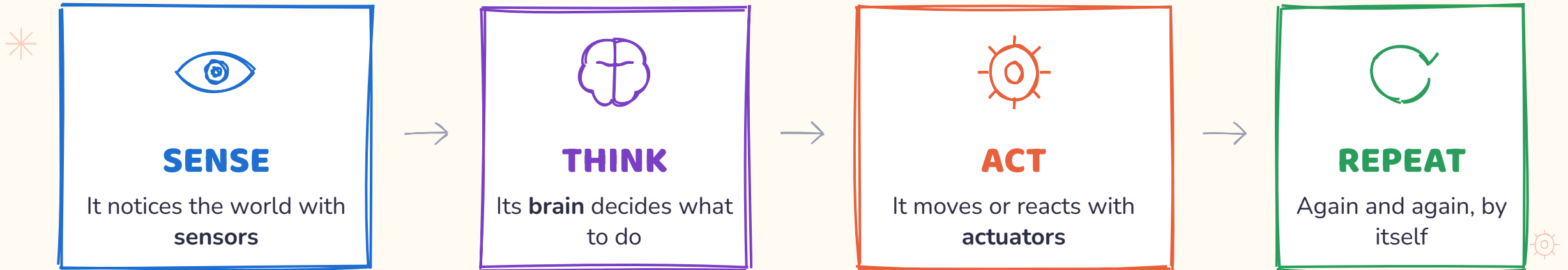


Talk with your partner: **what makes something a robot** and not just a machine?



The big idea

A robot can SENSE, THINK and ACT



Our definition

A robot is a **machine** that can **sense** its surroundings, **decide** what to do using a **program**, and then **act**, all **by itself**.

You do it too!

You **SEE** a ball coming, your brain **THINKS** 'catch!', your hands **ACT**.



Answers



So... which ones are robots?

Sort of**Washing machine**

It follows a program and senses water level, but it cannot move around or change plans much.

No**Remote-control car**

A person does all the thinking. No sensing, no deciding by itself.

Yes!**Robot vacuum**

Senses walls and dirt, decides where to go, moves by itself.

No**Calculator**

It thinks (calculates) but cannot sense the world or move.

Yes!**Self-driving car**

Cameras and radar sense, a computer decides, motors steer and brake.

Yes!**Delivery drone**

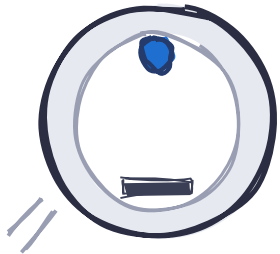
Uses GPS and sensors to fly a route and land on its own.

The more a machine can **sense, decide and act on its own**, the more "robot" it is. Engineers call this being **autonomous**.



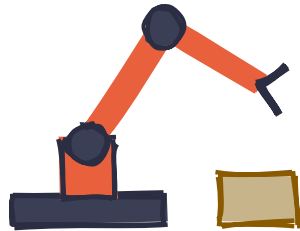
Look around you

Robots are everywhere!



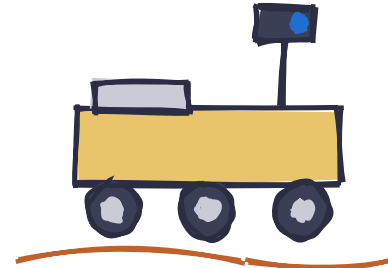
Home helper

Cleans floors by itself



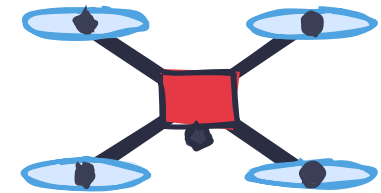
Factory arm

Builds cars, packs boxes



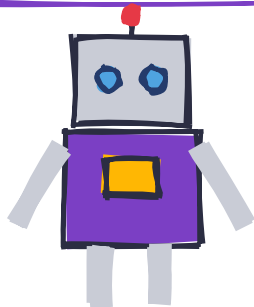
Mars rover

Explores another planet



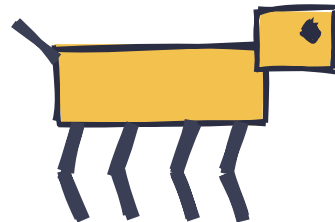
Drone

Films, maps, delivers



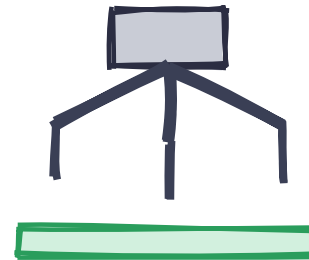
Humanoid

Walks and talks like us



Robot dog

Inspects dangerous places



Surgical robot

Helps doctors operate



Self-driving car

Drives itself on roads



A short history

From an idea to everywhere



1921

The word **robot** first appears in a Czech play by Karel Čapek. It comes from *robota*, meaning hard work.

1997

Sojourner, a rover the size of a microwave, drives on Mars.

2021

Ingenuity makes the first powered flight on another planet, Mars.

1961

Unimate, the first factory robot arm, starts work in a car factory in the USA.

2002

Roomba robot vacuums go on sale and robots enter our homes.

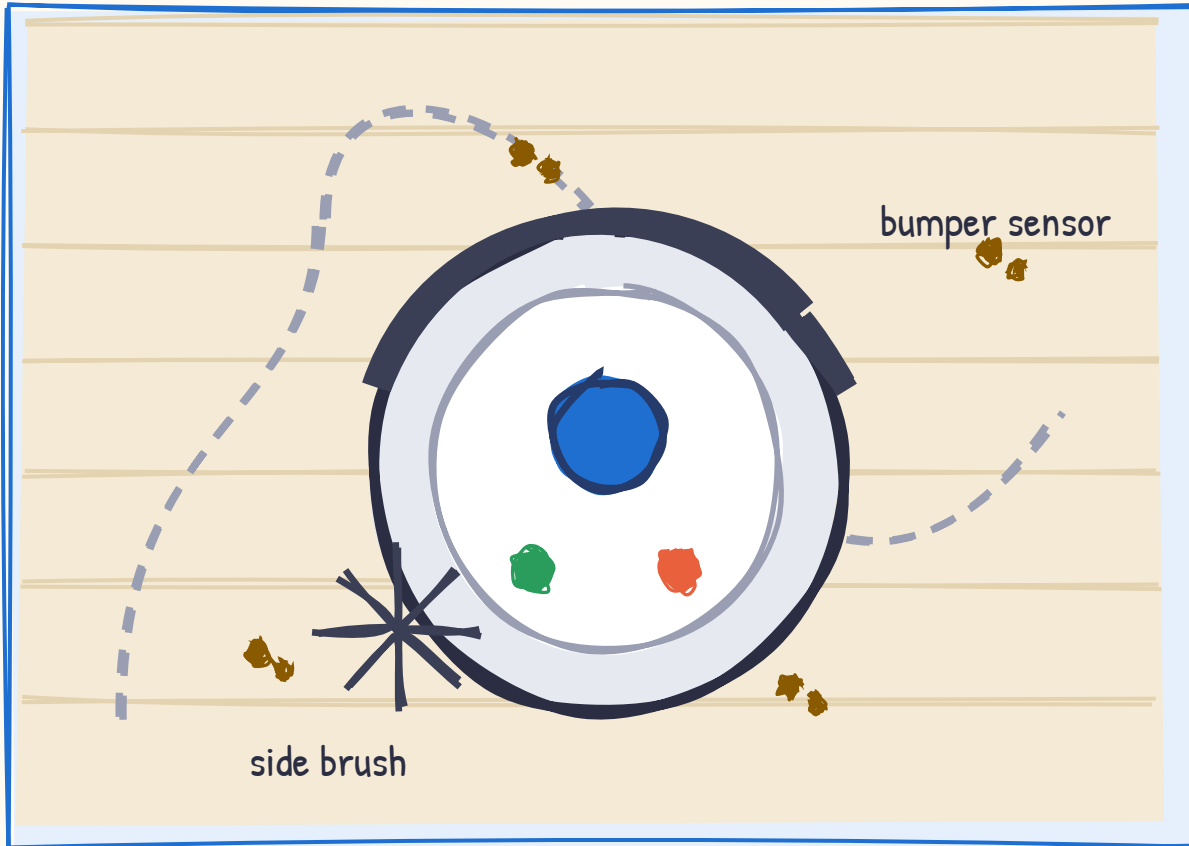
Today

Robots work in hospitals, farms, warehouses and space, often with **AI** inside.



Robot of the home · iRobot, USA · 2002

Roomba



What it does

A round robot that **vacuums your floor by itself**. It drives around furniture, cleans under beds and goes back to its **charging dock** when its battery is low.

SENSE: bump sensors, cliff sensors (stop falls down stairs), dirt sensors

THINK: plans a cleaning path

ACT: wheels, spinning brushes and a vacuum

Fascinating fact

iRobot has sold tens of millions of robot vacuums. Many homes have a robot before they have any other robot!



Robot dog · Boston Dynamics, USA

Spot



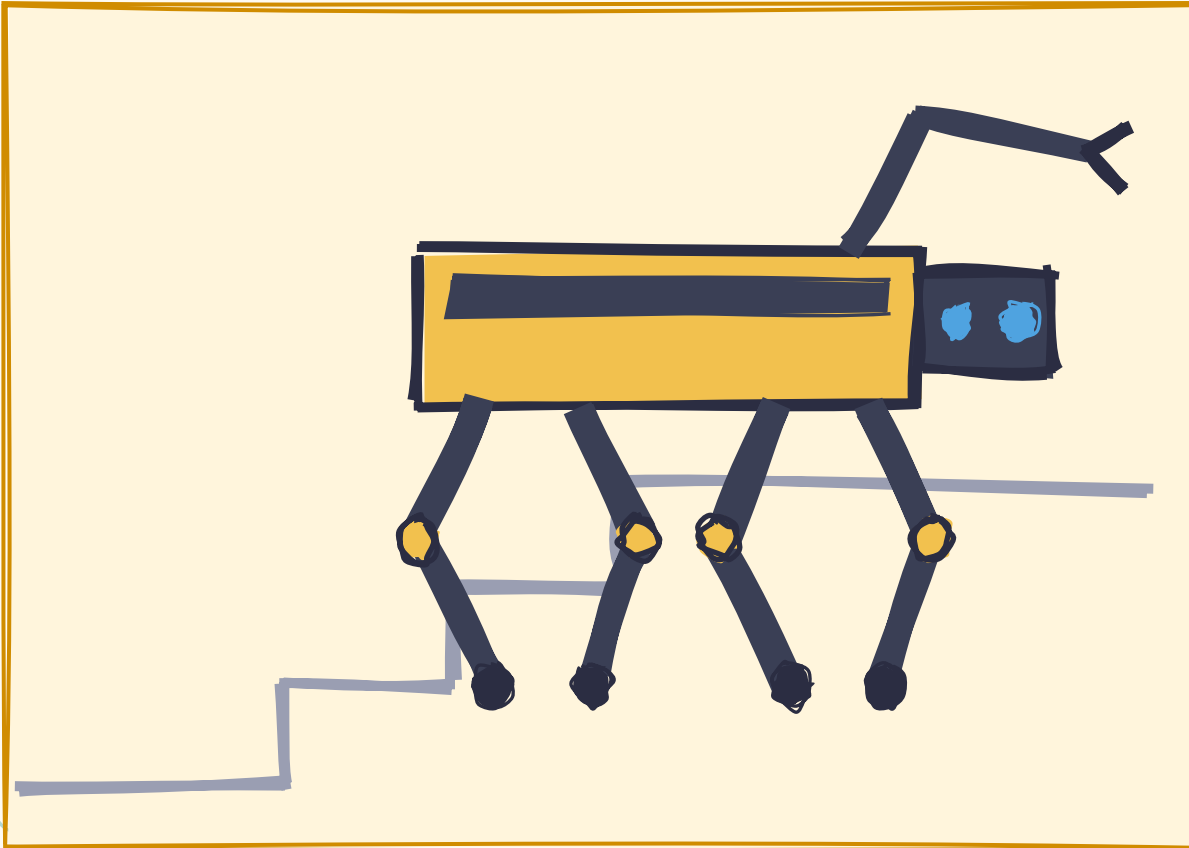
What it does

A four-legged robot that **walks, climbs stairs and opens doors**. It inspects places that are dangerous for people, like power stations, oil rigs and building sites.

SENSE: cameras on every side see in 3D all around it
THINK: keeps its balance hundreds of times every second
ACT: 12 motors move its 4 legs; an optional arm can grab things

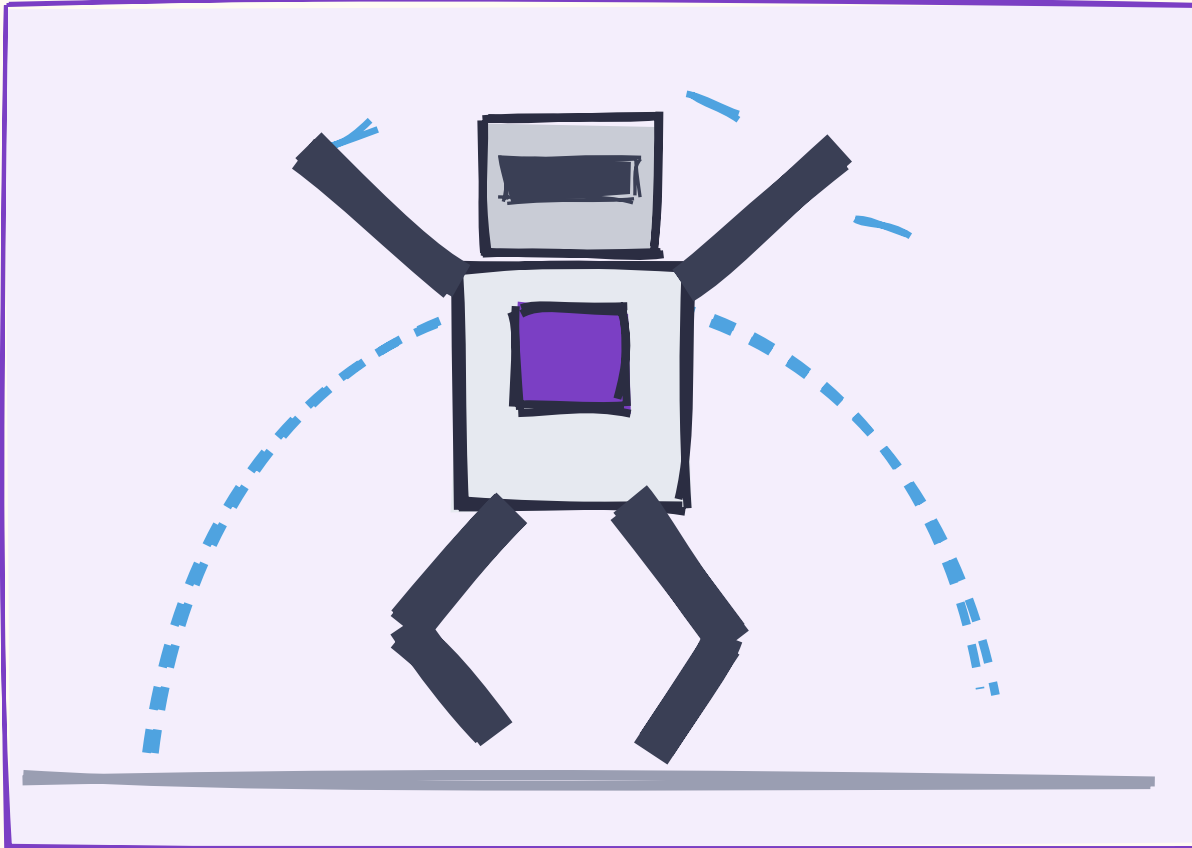
Fascinating fact

Spot can carry about 14 kg on its back and walk on rough ground where wheels would get stuck.



Humanoid robot • Boston Dynamics, USA

Atlas



What it does

A robot shaped like a **human** that can run, jump and even do **backflips**. Engineers use it to learn how robots can work in places built for people.

SENSE: cameras and depth sensors map the ground in front of it

THINK: a computer plans every jump and landing

ACT: powerful motors in its joints move like our muscles

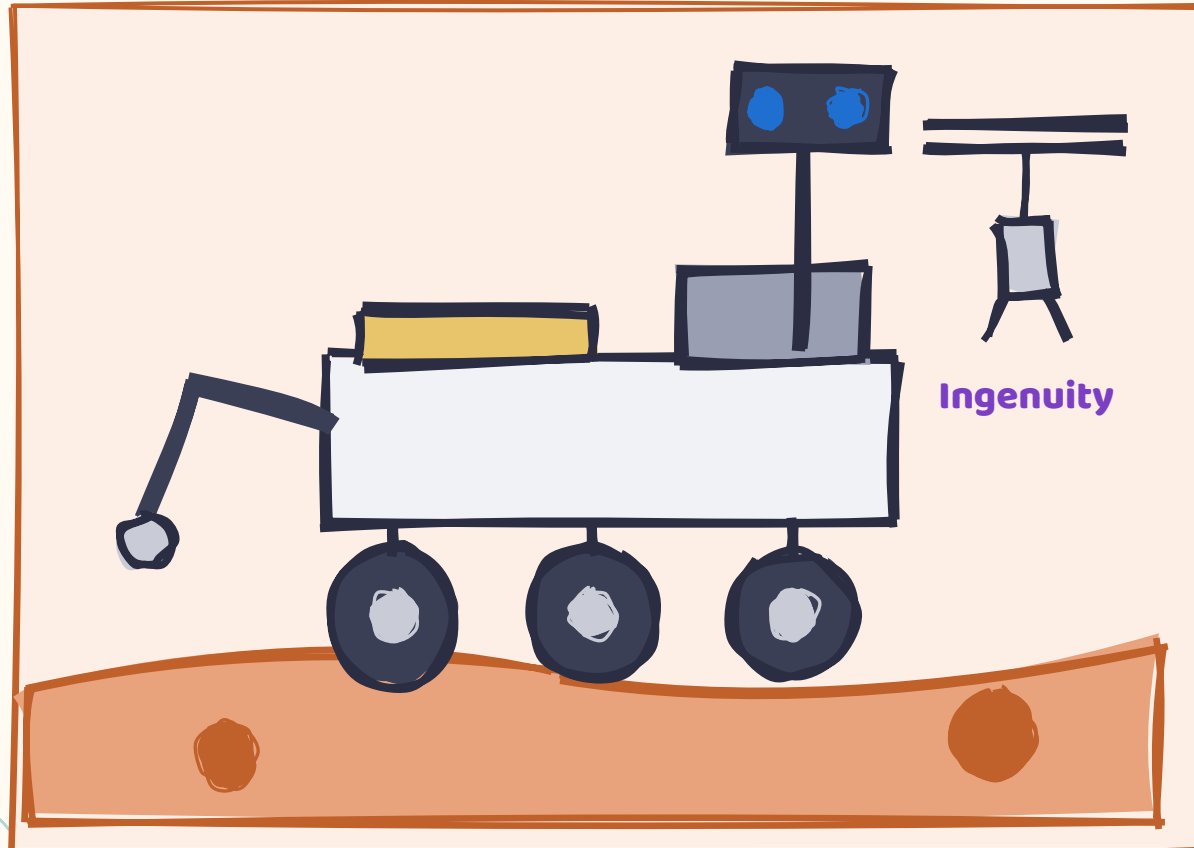
Fascinating fact

Atlas videos doing parkour have been watched by millions of people. Balancing on two legs is one of the hardest problems in robotics!



Mars rover · NASA, USA · landed 2021

Perseverance



What it does

A car-sized robot **exploring Mars**. It drills rocks, collects samples and searches for signs of ancient life. It also carried **Ingenuity**, a small helicopter that flew 72 times on Mars.



SENSE: about two dozen cameras, a laser and science instruments

THINK: drives itself between targets, because radio messages take minutes to reach Earth

ACT: 6 wheels, a robotic arm with a drill, and sample tubes

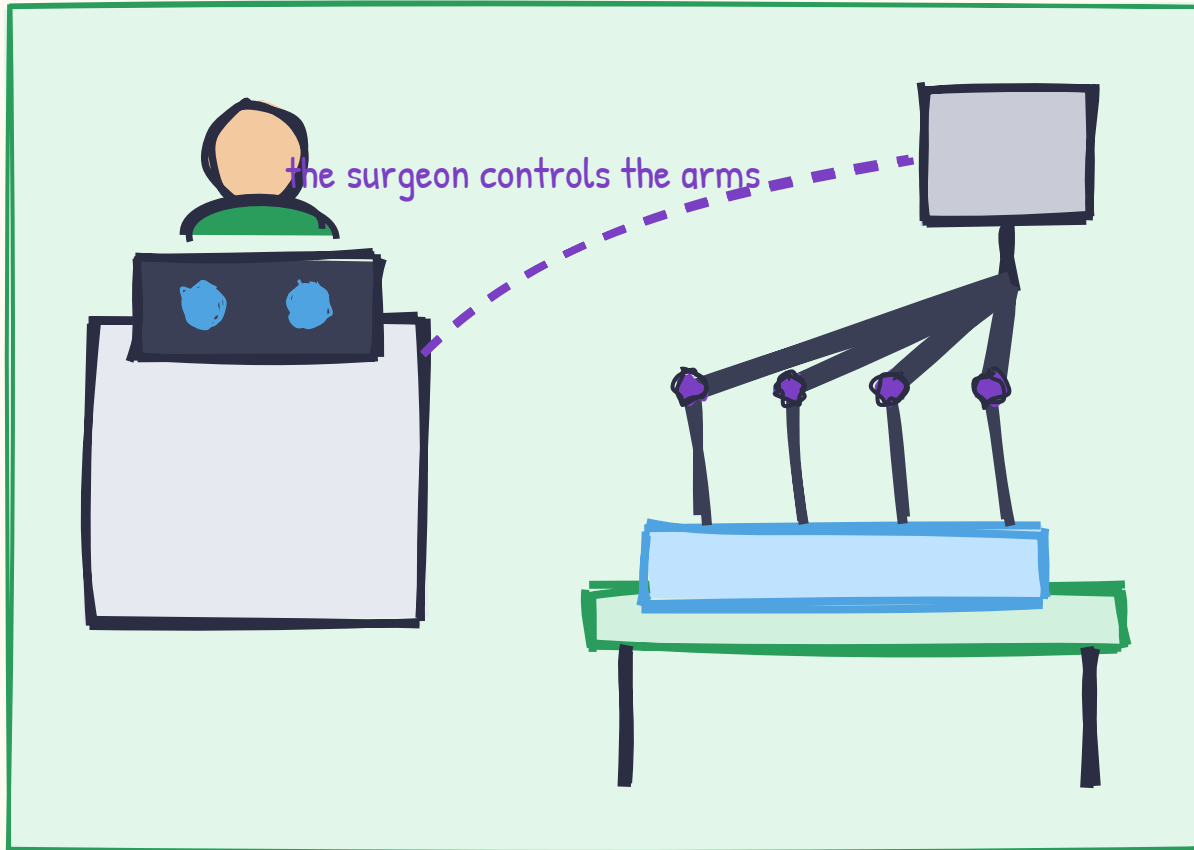
Fascinating fact

Perseverance weighs about 1,000 kg and is powered by a nuclear battery, not solar panels.



Surgical robot · Intuitive, USA

da Vinci



What it does

A robot that **helps doctors do operations** through tiny cuts. The surgeon sits at a console and moves controls; the robot's arms copy the movements with tiny tools.

SENSE: a 3D camera inside the body gives the surgeon a magnified view

THINK: turns big hand movements into tiny precise ones and filters out shaking

ACT: 4 robotic arms hold the camera and the instruments

Fascinating fact

Robots like this have helped with millions of operations around the world. The surgeon is always in control.



Social humanoid · Hanson Robotics, Hong Kong · 2016

Sophia



What it does

A robot with a **human-like face** that can smile, frown and hold simple conversations. She shows how robots might talk with people in the future.

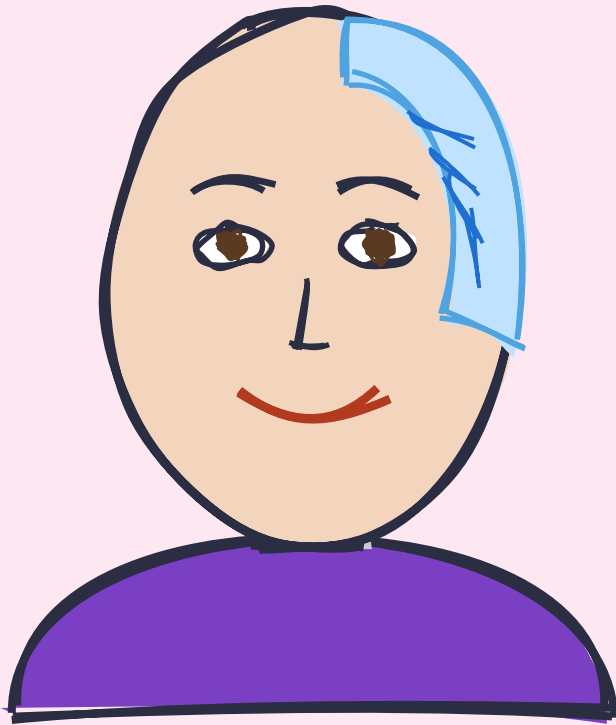
SENSE: cameras in her eyes see faces; microphones hear speech

THINK: AI software chooses what to say and which expression to show

ACT: small motors under her skin move her face

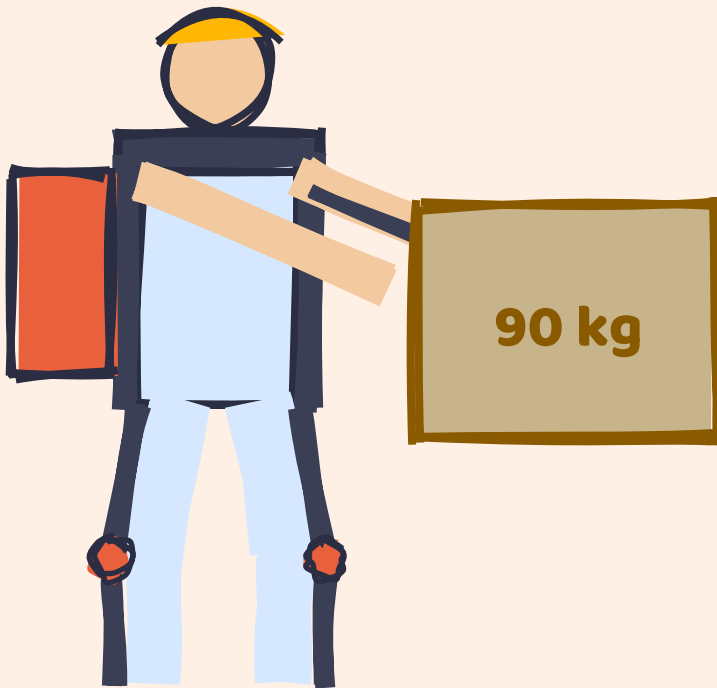
Fascinating fact

In 2017 Sophia became the first robot to be given citizenship of a country (Saudi Arabia).



A wearable robot · real-life super suit

Exoskeleton



What it does

A robot you **wear**, like a metal skeleton outside your body. Factory workers use them to lift heavy loads, and medical ones help people who cannot walk to stand and walk again.

SENSE: sensors feel how the wearer moves and how heavy the load is

THINK: decides how much extra force to add, many times a second

ACT: motors at the hips, knees and shoulders add strength

Fascinating fact

Some industrial exoskeletons let a worker lift around 90 kg again and again, with the robot taking most of the weight.



Moving robots • e.g. Amazon fulfilment centres

Warehouse robots



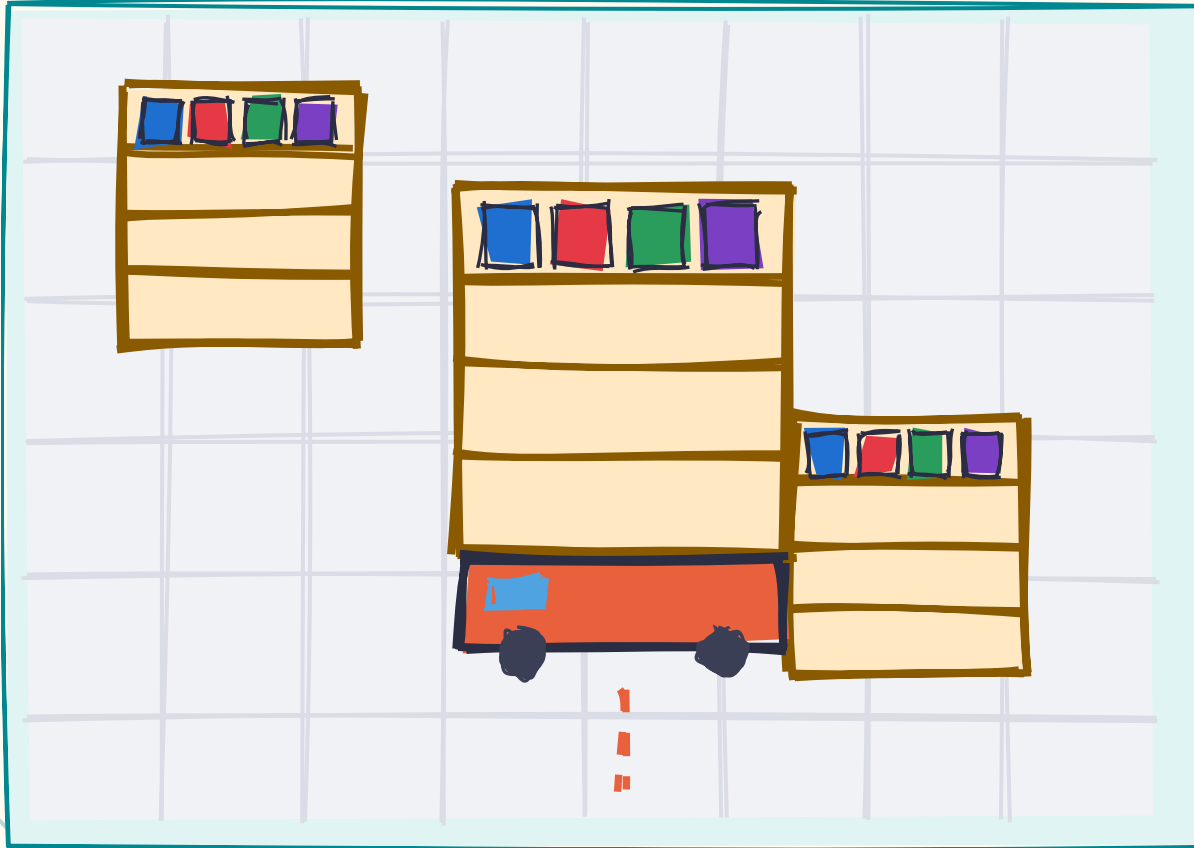
What it does

Hundreds of small robots **drive under shelves**, **lift them and carry them** to workers who pack orders. They zoom around without crashing into each other.

SENSE: cameras read codes stuck on the floor to know exactly where they are
THINK: a central computer plans routes for every robot at once
ACT: wheels drive them; a lifting plate raises the shelf

Fascinating fact

Amazon uses around one million robots in its warehouses, working side by side with people.



Inside every robot

The 6 parts of a robot



Sensors = the eyes, ears and skin
Collect information



Controller = the brain
Runs the program, makes decisions



Actuators = the muscles
Move, light up, make sound



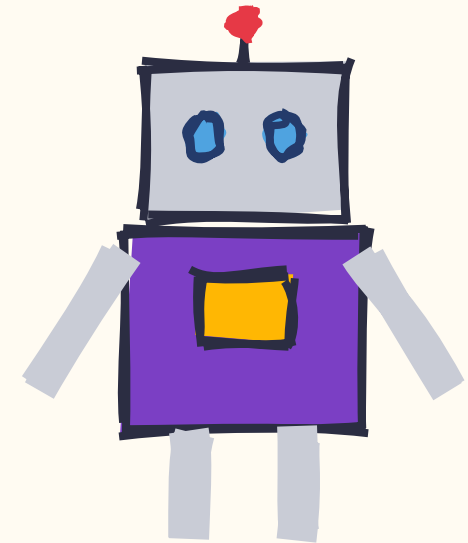
Power = the food and energy
Batteries or a power supply



Body = the skeleton
Holds everything together



Program = the thoughts
Instructions written by us



Human vs robot

We have the same 6 parts! Our nerves are the wires.



Sensing



Sensors: how robots notice the world

**Ultrasonic**

Distance, using sound echoes

**Light (LDR)**

How bright or dark it is

**Camera**

Pictures, faces, colours

**Touch / button**

Being pressed or bumped

**Temperature**

How hot or cold it is

**Motion (PIR)**

A person moving nearby



Today we use the **ultrasonic sensor**. It shouts a sound too high for us to hear and listens for the **echo**, just like a bat.

Your sensors

Eyes, ears, nose, tongue, skin. Which one is like a camera?





Acting

Actuators: how robots move and react

**DC motor**

Spins wheels and fans

**Servo motor**

Turns to an exact angle

**LED**

Lights up to show a signal

**Buzzer**

Beeps an alarm or a tune

**Screen**

Shows words and numbers

**Pump**

Moves water



Today our robot uses two actuators: a **red LED** to warn your eyes and a **buzzer** to warn your ears.

Think

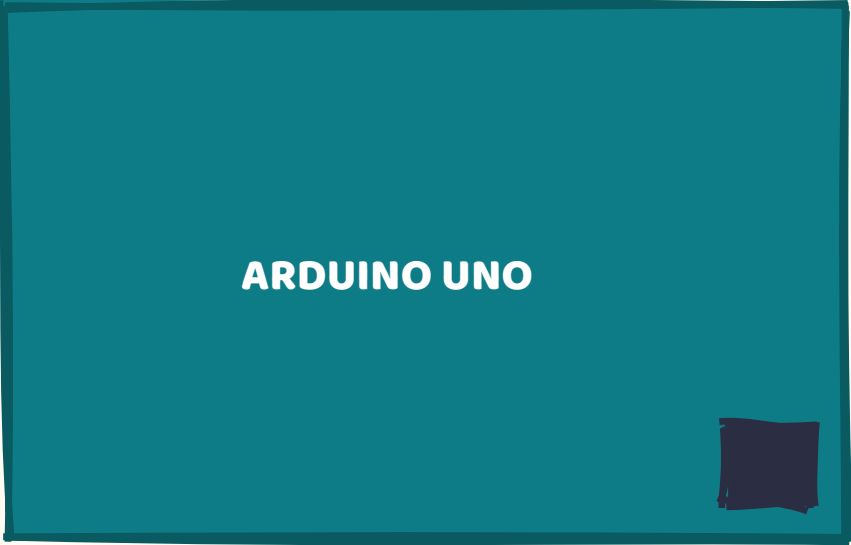
Which actuators does a robot vacuum need?





Thinking

The brain: a microcontroller



ARDUINO UNO

Bigger brain

Later we use the ESP32: a brain with Wi-Fi and Bluetooth built in!

What is it?

A **microcontroller** is a tiny computer on a board. The **Arduino Uno** reads sensors, runs our program and controls actuators. It is cheap, safe (5 volts) and used by millions of learners.



How do we program it?

1. Write the program on a laptop.
2. **Upload** it through a USB cable.
3. The Arduino runs it **by itself**, even without the laptop.



Robots + AI

When robots learn

A normal robot

Follows **rules we write**.

IF something is closer than 20 cm, THEN beep.

A robot with AI

Learns from **examples**.

Show it 1,000 pictures of cats and it learns to *recognise* a cat by itself.



Face unlock on phones



Cars that read road signs



Farm robots that pick ripe fruit



Voice assistants that understand you



Stand up and play!

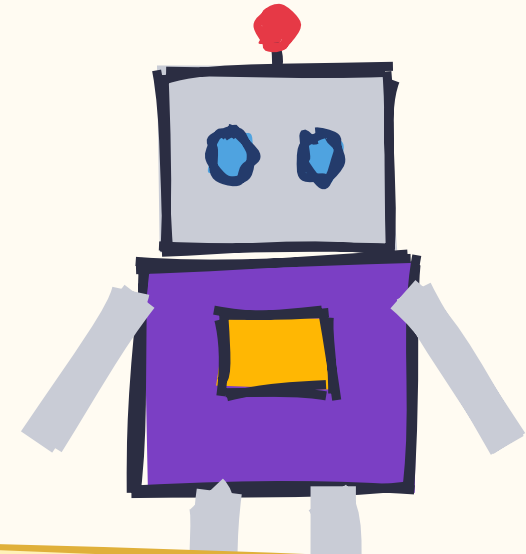
Program a human robot

How to play (pairs)

1. One partner is the **robot**, one is the **programmer**.
2. The robot must reach a target (a chair, a door) and pick up an object.
3. The robot follows instructions **exactly**, nothing more: "step forward", "turn right", "stop".
4. If the program is wrong, the robot does the wrong thing!



Was your program right the first time? Fixing mistakes in a program is called **debugging**.



Robot rule

Robots do what we TELL them, not what we MEAN!



Your first build

Make a Distance Alarm robot



Like a car parking sensor

When a car reverses towards a wall, it **beeps** faster and faster. Today you build the same thing: when something comes **closer than 20 cm**, a red light turns on and a buzzer beeps.

Sense, think, act

Sense: ultrasonic sensor measures distance

Think: Arduino checks: closer than 20 cm?

Act: LED on + buzzer beeps



Build it

Our builder's kit (per pair)



Arduino Uno + USB

The brain



HC-SR04 sensor

The eyes (ultrasonic)



Red LED + 220 Ω

Light alarm



Active buzzer

Sound alarm



Breadboard + wires

Roads for electricity



Laptop

Upload the program



Two ways to build

Ready-built: the teacher has wired it. Plug in, upload, test and change the program.

Build it yourself: follow the wiring guide on the next slide, then ask the teacher to check before you plug in.

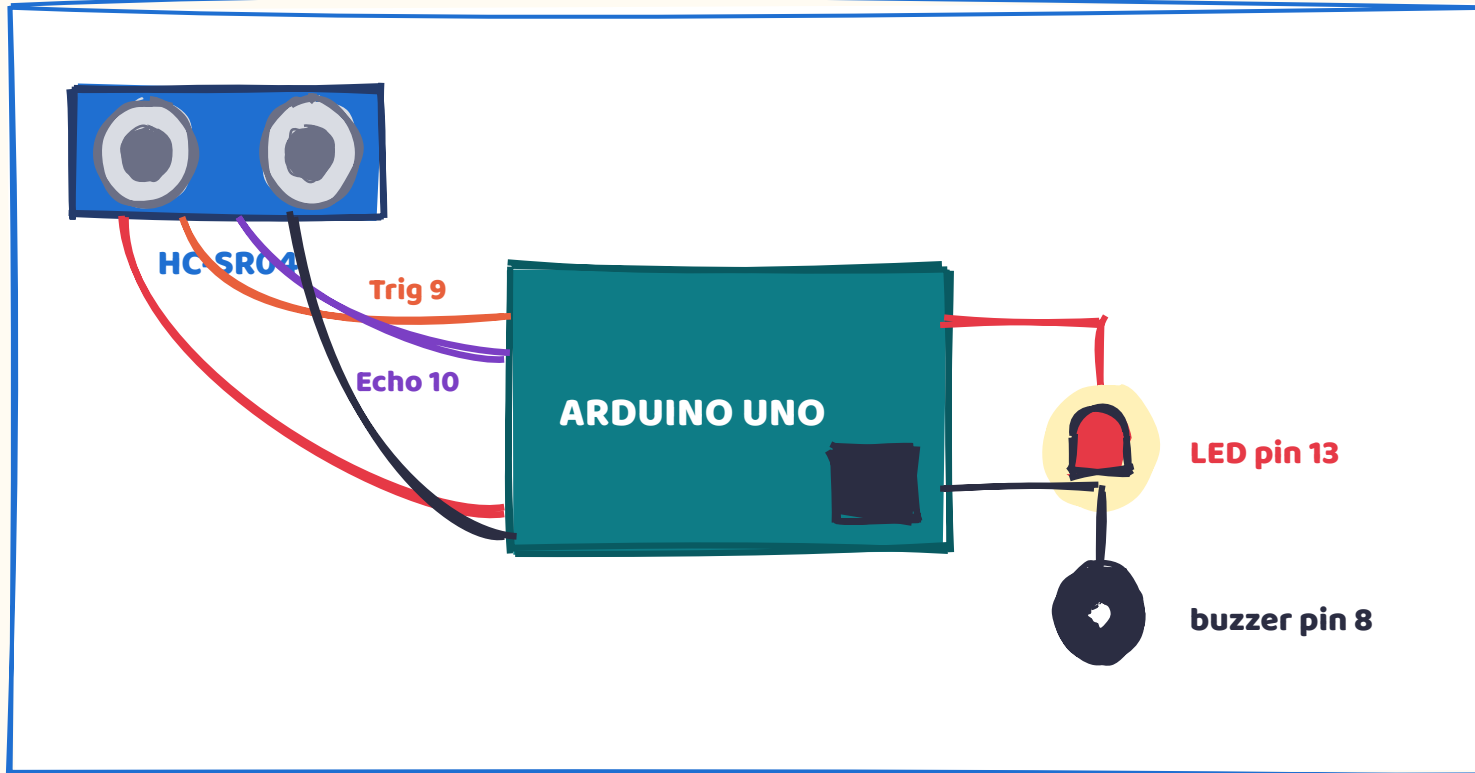
Safety first

Always unplug the USB cable before you change any wires.



Build it

Wiring guide



Part	Arduino
Sensor VCC	5V
Sensor Trig	pin 9
Sensor Echo	pin 10
Sensor GND	GND
LED long leg (+ 220 Ω)	pin 13
Buzzer +	pin 8
LED short leg, buzzer -	GND

Check twice

Red wire to 5V, black wire to GND. Swapping them can damage the sensor!



The program

Ready-made code: just upload!



```
// Distance Alarm: like a car parking sensor
const int TRIG = 9, ECHO = 10, LED = 13, BUZZ = 8;
int ALARM_CM = 20;           // <-- change me!

void setup() {
  pinMode(TRIG, OUTPUT);  pinMode(ECHO, INPUT);
  pinMode(LED, OUTPUT);   pinMode(BUZZ, OUTPUT);
  Serial.begin(9600);
}

void loop() {
  digitalWrite(TRIG, HIGH); delayMicroseconds(10);
  digitalWrite(TRIG, LOW);           // SENSE: send a sound pulse
  long cm = pulseIn(ECHO, HIGH, 30000) / 58;
  Serial.println(cm);                // see the distance

  if (cm > 0 && cm < ALARM_CM) {      // THINK: too close?
    digitalWrite(LED, HIGH);         // ACT: light on
    digitalWrite(BUZZ, HIGH);        // and beep
  } else {
    digitalWrite(LED, LOW);
    digitalWrite(BUZZ, LOW);
  }
  delay(100);
}
```

Find the 3 parts

SENSE: send a sound and time the echo.

THINK: if the distance is less than ALARM_CM...

ACT: turn the LED and buzzer on.



Your only job

Change ALARM_CM from 20 to another number. Upload again. What changes?

Teacher: the file is shared on the lab laptops. Open the Serial Monitor (9600) to see live distances.



Test like an engineer

Does your robot sense correctly?



Test	Expect	Reading	Worked?
Hand 50 cm away	LED off, quiet	_____ cm	yes / no
Hand 10 cm away	LED on, beeping	_____ cm	yes / no
A book instead of a hand	Does it still work?	_____ cm	yes / no
A soft cushion or sweater	Does it still work?	_____ cm	yes / no

Not working?

Always 0? Trig and Echo are swapped.
LED never lights? Long leg to pin 13?
No beep? Buzzer + to pin 8?
Nothing at all? Is the USB plugged in?



Why might a soft cushion fool the sensor? (Hint: what happens to sound when it hits something soft?)



Level up!

Choose your challenge



Level 1

Change the alarm distance to 10 cm, then to 50 cm.



Level 2

Add a green LED that is ON when everything is safe.



Level 3

Make it beep faster as things get closer, just like a real car.

```
// Level 3 hint
digitalWrite(BUZZ, HIGH);
delay(50);
digitalWrite(BUZZ, LOW);
delay(cm * 10); // closer = shorter gap
```



Real-world idea

Where else could a distance alarm help? A blind person's cane? A garage door?





Coming up in the club

What will we build next?



Robot cars

that avoid walls and follow lines



Robot arms

that copy your hand



Smart devices

you control from your phone



Smart farms

that water plants by themselves



Safety systems

for fire, gas and security



AI projects

that see and recognise things



Robotics jobs: **robotics engineer**, **AI engineer**, **drone pilot**, **electronics designer**, **programmer**... and jobs that do not even exist yet!





Show what you know

What did we learn?



- ☒ A robot senses, thinks and acts by itself
- ☒ Robots have sensors, a controller, actuators, power, a body and a program
- ☒ Robots work in homes, factories, hospitals and even on Mars
- ☒ An ultrasonic sensor measures distance with echoes
- ☒ A program tells the robot exactly what to do



Robot Builder

badge unlocked!



At home

Count the robots and sensors you can find at home.
Bring your list next class!



Exit ticket: name one SENSOR and one ACTUATOR in your Distance Alarm.

